

Kolmogorov–Arnold Networks: Overview of Architectures and Use Cases

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Abstract—Kolmogorov–Arnold Networks (KANs) are an emerging class of neural network architectures grounded in the Kolmogorov–Arnold representation theorem, offering a symbolic and interpretable approach to modeling complex, high-dimensional functions. By replacing traditional activation functions with learnable univariate splines, KANs deliver enhanced expressiveness, parameter efficiency, and transparency. This survey provides a comprehensive overview of KAN developments, including foundational designs and specialized variants tailored to time series, graphs, scientific computing, image analysis, and biomedicine. We highlight the strengths of KANs in bridging the gap between symbolic reasoning and data-driven learning, while also discussing challenges such as computational overhead and spline optimization complexity. Through detailed architectural taxonomy and real-world application examples, this paper positions KANs as a compelling framework for interpretable, high-performance machine learning.

Index Terms—Kolmogorov–Arnold Networks, Deep Learning, KANs, Machine Learning.

I. INTRODUCTION

KANs [1], [2] are a class of neural network architectures inspired by the Kolmogorov–Arnold representation theorem, which states that any multivariate continuous function can be expressed as a finite superposition of univariate continuous functions [4]. Unlike traditional neural networks [13] that rely on fixed activation functions and weight matrices [12], KANs replace these with learnable, spline-parameterized functions, offering a more flexible and interpretable alternative for modeling high-dimensional nonlinear functions.

KANs have emerged at the intersection of theoretical mathematics and deep learning, leveraging the rigor of functional decomposition to offer an architectural alternative that is not only expressive but also inherently interpretable. Their formulation enables neural models to learn continuous mappings across dimensions through the use of trainable univariate functions, often constructed using B-splines, wavelets, or orthogonal polynomials [39], [42]. This contrasts with traditional MLPs that rely on fixed nonlinearities such as ReLU or tanh [27], making KANs naturally suited for applications where model behavior must be understood and trusted.

In recent years, there has been growing interest in interpretable and parameter-efficient machine learning models [28], particularly in domains where transparency and domain alignment are critical—such as biomedicine [3], [29], physics [24], and financial modeling [14], [25], [26]. KANs, by design, offer symbolic insights into learned mappings, enabling both performance and interpretability. They present a viable solution to the “black-box” nature of deep learning [38] by modeling neural computation through mathematical decompositions grounded in classical function theory.

The renewed attention on KANs stems from both theoretical and practical advancements. New variants such as FastKAN [39], PDE-KAN [42], and T-KAN [5] have shown that KANs can match or exceed the performance of Multi-Layer Perceptrons (MLPs), Convolutional Neural Networks (CNNs) [27], and even Transformers [30], while using fewer parameters and maintaining transparency in learned representations. Their use of edge-based activation, smooth function modeling, and parameter sparsity allows for more data-efficient training and enhanced symbolic interpretability.

Moreover, the growing ecosystem of KANs variants now supports diverse applications such as symbolic regression, scientific computing, image processing [37], time series forecasting, and graph representation learning. These models have been shown to outperform baseline networks in both synthetic tasks and real-world settings, offering a strong case for further exploration and deployment in critical domains [64].

This paper makes the following contributions:

- Provides a comprehensive overview of KAN architectures, including foundational models and recent developments tailored to time series, graphs, physics-informed learning, and hybrid deep architectures.
- Presents an analysis of use cases where KANs demonstrate competitive advantages in terms of accuracy, efficiency, and interpretability.
- Highlights the theoretical underpinnings that make KANs a compelling alternative to traditional deep networks,

while discussing practical challenges and open research directions.

II. THEORETICAL FOUNDATIONS

The Kolmogorov–Arnold representation theorem [59], introduced by Andrey Kolmogorov and later refined by Vladimir Arnold, states that any continuous multivariate function $f(x_1, x_2, \dots, x_n)$ can be expressed as:

$$f(x_1, x_2, \dots, x_n) = \sum_{q=1}^{2n+1} \phi_q \left(\sum_{p=1}^n \psi_{q,p}(x_p) \right) \quad (1)$$

This theorem implies that a high-dimensional function can be approximated by composing and summing simpler, one-dimensional functions. KANs leverage this insight by constructing neural architectures where each edge represents a learnable univariate function, typically implemented as splines or other basis expansions (e.g., Chebyshev or Jacobi polynomials).

A. Comparison with Traditional Neural Networks

Traditional MLPs [60] compute outputs through compositions of affine transformations and fixed nonlinearities (see Figure 1):

$$h_i = \sigma \left(\sum_j w_{ij} x_j + b_i \right) \quad (2)$$

where σ is a non-linear activation function and w_{ij} are scalar weights. The expressiveness of MLPs is tied to the depth of the network and the shape of σ , limiting their flexibility in modeling fine-grained, local function behaviors.

In contrast, KANs compute activations using learnable univariate functions:

$$h_i = \sum_j \phi_{ij}(x_j) \quad (3)$$

where ϕ_{ij} is a data-driven, smooth function trained during learning. This yields several benefits:

- **Expressiveness:** KANs can model complex nonlinearities more naturally and with fewer layers or parameters.
- **Interpretability:** The shape and behavior of each ϕ_{ij} can be directly inspected, providing insights into the model’s decision process.
- **Theoretical Foundation:** KANs directly implement a known functional decomposition, offering principled approximations beyond empirical generalization.

This structured, interpretable framework positions KANs as a compelling alternative for applications that require both accuracy and transparency.

III. OVERVIEW OF KOLMOGOROV–ARNOLD NETWORKS ARCHITECTURES AND USE CASES

KANs have undergone significant architectural evolution, expanding from their theoretical roots into numerous specialized and hybrid forms. These variants are designed to address domain-specific challenges while retaining the core philosophy of adaptive univariate functional decomposition. Below, we provide a comprehensive overview of KAN architectures based on recent advancements [1], [2], [4].

A. Core and Foundational Models

KANs [1] is the original architecture that employs spline-based univariate activation functions on the edges of the network graph instead of fixed nonlinearities at nodes, enabling flexible and interpretable function approximation. It has been used in symbolic regression and low-dimensional scientific problems, and its edge-based activation scheme allows the model to learn continuous mappings, making it well-suited for tasks requiring high interpretability. FastKAN [39] is an optimized variant that replaces computationally expensive B-splines with Gaussian Radial Basis Functions (RBFs), significantly accelerating training while maintaining similar performance. It has shown strong results on tabular datasets and is ideal for rapid prototyping. Spline-KAN¹ generalizes the base model by allowing a variety of spline bases (e.g., cubic, B-spline), increasing adaptability across domains. BSRBF-KAN [40] merges B-splines and RBFs to achieve greater function approximation in high-dimensional input spaces, often applied in physical simulation environments. fKAN [41] incorporates fractional Jacobi Neural Blocks, using parameterized Jacobi polynomials to represent symbolic functions and is particularly useful in modeling physical processes governed by PDEs.

B. Time-Series and Sequential Models

T-KAN [5] is tailored for univariate time series forecasting using sliding windows and has demonstrated high predictive accuracy on solar power, temperature, and wind datasets. MT-KAN [5] extends this approach to multivariate inputs, allowing shared representations across variables and achieving success in domains such as smart grid forecasting. TKANs augment the KAN layer with LSTM-style memory gates, making them capable of capturing long-range dependencies in temporal sequences while preserving interpretability. SigKAN [6] incorporates path signature features and gated residual blocks, enhancing its ability to model irregular time series data like medical records or multi-modal biosignals. BiLSTMKAN-net [7] combines bidirectional LSTM cells with DenseKAN layers, offering bi-directional temporal insight and has been used in anomaly detection for industrial systems. HiPPO-KAN [8] uses HiPPO memory representations to capture continuous-time dynamics efficiently, outperforming standard RNNs on memory-intensive benchmarks. Quantum-based QCKANet [7], QDenseKANnet [7], and QKAN [9] use hybrid quantum-classical computation frameworks and have been proposed for

¹https://github.com/pg2455/KANTutorial/blob/main/1_splines.ipynb

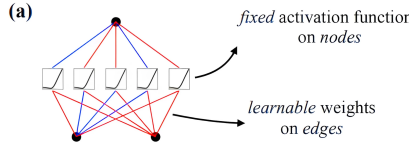
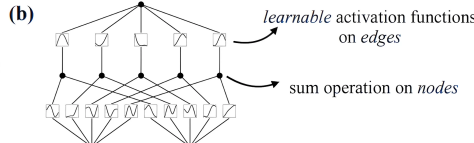
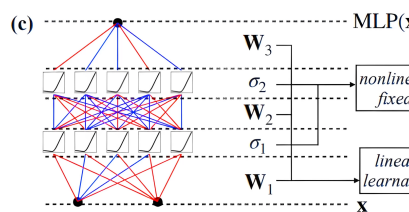
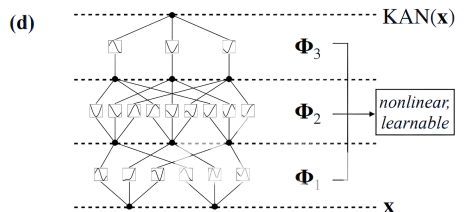
Model	Multi-Layer Perceptron (MLP)	Kolmogorov-Arnold Network (KAN)
Theorem	Universal Approximation Theorem	Kolmogorov-Arnold Representation Theorem
Formula (Shallow)	$f(\mathbf{x}) \approx \sum_{i=1}^{N(\epsilon)} a_i \sigma(\mathbf{w}_i \cdot \mathbf{x} + b_i)$	$f(\mathbf{x}) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^n \phi_{q,p}(x_p) \right)$
Model (Shallow)	(a) 	(b) 
Formula (Deep)	$\text{MLP}(\mathbf{x}) = (\mathbf{W}_3 \circ \sigma_2 \circ \mathbf{W}_2 \circ \sigma_1 \circ \mathbf{W}_1)(\mathbf{x})$	$\text{KAN}(\mathbf{x}) = (\Phi_3 \circ \Phi_2 \circ \Phi_1)(\mathbf{x})$
Model (Deep)	(c) 	(d) 

Fig. 1. Comparison between MLP and KAN [1].

tasks like quantum state discrimination and low-dimensional manifold learning.

C. Scientific Computing and Symbolic Models

PDE-KAN [42] is a physics-informed model used to solve boundary value problems like Poisson and Helmholtz equations. Its architecture embeds domain knowledge directly into the loss function and has demonstrated mesh-free numerical stability. PIKAN [21] relies on polynomial expansions to represent differential operators and is effective in settings with limited or noisy measurements. KAN2CD [43] adopts a hierarchical structure that maps raw inputs to conceptual features and has been tested in student performance modeling and intelligent tutoring systems. KINN [42] integrates structured priors derived from physics or domain heuristics, showing promise in modeling phenomena like turbulent fluid flow. EPi-cKAN [44] employs Chebyshev polynomial expansions and enables the simulation of mechanical stress responses, making it applicable in civil and aerospace engineering. rKAN [45] uses rational functions to approximate sharp transitions or discontinuities, making it ideal for scientific tasks requiring symbolic regression or logic-based inference.

D. Graph Learning and Structured Data

GraphKAN [10] leverages spline-based edge functions to process graph-structured inputs and has excelled in few-shot classification tasks on datasets such as Cora, Citeseer, and PubMed. GKAN [11] proposes two distinct designs: one integrates KAN after graph aggregation while the other applies it before message passing. Both show improvements in node and edge classification accuracy. KA-GNN [15], KAGCN [16], and KAGIN [16] replace traditional dense layers with

KAN modules in graph neural networks and have been used in cheminformatics for molecular property prediction and in transportation systems for dynamic routing. PointNet-KAN [17] extends the PointNet architecture using KAN layers with Jacobi polynomial activations, improving robustness and accuracy in 3D object classification from point clouds.

E. Image and Spatial Models

KANICE [46] integrates KAN into CNN pipelines using Interactive Convolutional Blocks to enhance model robustness against adversarial perturbations, demonstrating superior performance on CIFAR-10 and MNIST. C-KAN [47] fuses convolutional filters with spline-based nonlinearities, making it effective for spatiotemporal modeling tasks like climate forecasting. RKAN [48] embeds Chebyshev polynomials into residual networks and has been successfully used in image denoising, compression artifact removal, and super-resolution. KAN-Mixer [50] serves as a lightweight image classification model, incorporating token mixing strategies with KAN nonlinearities to match the accuracy of heavy architectures like ResNet with fewer parameters.

F. Hybrid and Specialized Architectures

MultKAN [2] enhances the expressive power of KANs by incorporating multiplicative computation nodes, which allows the modeling of complex polynomial expressions and interaction terms, and is particularly suited for symbolic reasoning benchmarks. Wav-KAN [18] replaces traditional activations with wavelet functions like DoG and Haar, capturing local and frequency-aware patterns, useful in biomedical and speech signal processing. S-KAN [19] introduces switchable activation pathways to select the most relevant function per

task, helping to generalize across distribution shifts. DropKAN [20] applies stochastic masking after activation, functioning as a dropout mechanism to prevent overfitting in small-data scenarios. DEEPOKAN [21] stacks multiple KAN layers and has demonstrated utility in high-dimensional feature spaces such as hyperspectral imagery. MFKAN [22] allows fusion of low- and high-fidelity simulation data by integrating fidelity-specific layers and is used in aerospace design optimization. S-ConvKAN [19] integrates KAN units within CNNs and has been validated in optical character recognition and other structured visual inference problems.

G. Transformer-Compatible and Advanced Models

SCKansformer [51] combines SCConv attention mechanisms with KAN-based decoders, allowing localized and global representation learning simultaneously, which has been validated on translation and summarization benchmarks. KANSformer [52] utilizes a Transformer encoder with KAN modules in the decoder for tasks like document classification and time series analysis, improving performance on tasks with long-term dependencies. Kaninfradet3D [53] is a 3D vision model that fuses KAN with cross-attention modules, enabling accurate scene understanding from 3D point clouds and multi-view data. Lightweight variants such as LSin-SKAN [54], LCos-SKAN [54], LArctan-SKAN [54], and LSS-SKANs [55] implement parameter-efficient spline functions tailored for real-time applications on edge hardware.

H. Biomedicine and Domain-Specific Models

KAN-EEG [23] is tailored for neurophysiological signal processing and has achieved competitive accuracy in EEG seizure detection with fewer parameters, making it viable for mobile health devices. CaLMPHosKAN [56] incorporates domain-specific biological priors in phosphorylation site prediction and has been benchmarked on mass spectrometry data. COEFF-KANs [57] use chemical structure embeddings and KAN layers for molecular property estimation, relevant in drug discovery and QSAR modeling. WormKAN [58] is specialized for analyzing neural activity in *C. elegans*, reconstructing dynamic behavior patterns from neuron-level recordings with high fidelity.

IV. CHALLENGES AND RESEARCH OPPORTUNITIES

Despite their numerous advantages and demonstrated success across domains, KANs present several open challenges that must be addressed to enable broader adoption and drive future innovation. Their unique structure—while offering interpretability and flexibility—also introduces complexities not commonly seen in traditional deep learning models. In this section, we explore these challenges in depth and propose future directions that could enhance the practical utility of KANs.

A. High Computational Cost in High-Dimensional Settings

KANs model univariate transformations along network edges using spline or basis function representations. In high-dimensional input spaces, the number of such functions grows

significantly, leading to increased computational complexity both during training and inference [61]. Unlike MLPs, where weights are scalar and activation functions are fixed, KANs require the evaluation of piecewise-defined or polynomial-based functions at every edge. This can result in slower convergence and higher memory consumption. Current research is exploring efficient implementations using sparse spline representations [62] and low-rank approximations [66] to mitigate this cost. Furthermore, hardware-accelerated evaluations (e.g., FPGA or GPU-friendly spline kernels) [67] could provide scalable solutions in the future.

B. Training Complexity Due to Spline Optimization

Optimizing the spline-based activation functions in KANs poses unique challenges. These functions are not only learnable but must also maintain smoothness, continuity, and often monotonicity—constraints that complicate gradient-based learning. Initialization becomes critical: poor starting values for spline coefficients can lead to vanishing gradients or erratic function behavior [32]. Some variants, like FastKAN [39], address this by replacing splines with Gaussian RBFs for stability [31], while others introduce regularization terms that penalize roughness or impose sparsity. Effective training thus often requires domain-specific tuning, adaptive learning rate schedules, and customized loss terms that are not commonly used in conventional deep learning [72].

C. Generalization Limits on Noisy Data

KANs are particularly adept at fitting structured and smooth data, aligning closely with symbolic representations. However, their expressive capacity can backfire in noisy environments, where they risk modeling spurious patterns or overfitting small fluctuations in the input. Unlike neural networks with fixed activations that inherently smooth out noise, KANs may generate complex functions that capture it instead. Recent solutions such as DropKAN [20] introduce stochastic masking after activation to regularize the learned functions. Still, more research is needed on how to induce inductive biases in KANs that promote generalization—such as leveraging priors from symbolic mathematics, physics, or domain knowledge [69].

D. Integration with Other Architectures

Several hybrid models have successfully embedded KAN layers within convolutional or transformer-based architectures—such as KANICE [46] for image robustness and SCKansformer [51] for biomedical sequence modeling. Nevertheless, these integrations are ad hoc and lack standardized frameworks or APIs. Incorporating KANs into end-to-end architectures introduces challenges in gradient flow, weight sharing, and computational efficiency. In particular, adapting KANs to operate on tensor inputs [70], or aligning them with positional encodings and attention mechanisms [71], remains a technical barrier. Standardized modules, like PyTorch-compatible KAN layers, and plug-and-play design patterns could facilitate more widespread use [49], [63].

E. Future Directions and Opportunities

Looking forward, several promising directions stand out. First, hybrid neural-symbolic frameworks [33] could incorporate KANs as interpretable layers in otherwise opaque models, enabling explainable AI applications. Second, KANs are uniquely suited for physics-informed learning—embedding [34], [35] domain constraints into spline function spaces offers a principled way to simulate real-world phenomena. Third, symbolic integration of KAN models could enable the extraction of human-readable formulas, facilitating downstream analysis or theorem proving. Finally, accelerating spline evaluations using modern hardware (e.g., through vectorized or quantized representations) [36] could position KANs as efficient models for edge AI or real-time scientific computing.

By overcoming these technical and structural challenges, KANs have the potential to emerge as key tools in next-generation machine learning models that are not only powerful but also understandable, tunable, and aligned with the interpretability needs of science and industry.

V. CONCLUSION

In this paper, we provided a comprehensive overview of Kolmogorov–Arnold Networks, tracing their theoretical foundations to the Kolmogorov representation theorem and exploring the rich ecosystem of architectures that have emerged. From the core spline-based models to advanced hybrids with graph, sequence, and vision architectures, KANs demonstrate remarkable adaptability and have proven effective across a wide range of domains.

We discussed how KANs excel in combining interpretability with deep learning, enabling symbolic and data-driven reasoning within the same framework. Through real-world applications in scientific computing, biomedicine, graph learning, time series forecasting, and image analysis, KANs show promise in addressing some of the key limitations of traditional neural networks.

As KANs research continues to mature, real-world deployment will hinge on resolving current bottlenecks such as computational overhead and training complexity. Nonetheless, with increasing attention to symbolic AI, physics-informed learning, and hybrid systems, KANs offer a forward-looking architecture poised for wide applicability and enduring impact.

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